

## Aerospace & Defence, US Telecom & Communications Infrastructure

# SpaceX: Potential for MVNO D2D growth? Continuing our assessment of SpaceX Connectivity business and telco impact



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As SpaceX prepares for Starship launch 14, the company plans to deploy the first operational V3 Starlink satellites. In this note, we continue our focus on the Connectivity portion of the business and the approach we see as most attractive for direct-to-device service. We previously discussed the technical challenges of D2D ([SpaceX: The Telecom Play - Can Direct-to-Device Work?](#)) and the hurdles if the company were to build out its own terrestrial network ([SpaceX: More on Direct-to-Device Mobile - what would a buildout cost? Our detailed bottom-up estimates](#)). Here, we assess the potential for an MVNO option, which we have seen as most attractive since our initiation on SpaceX. **Despite historical challenges of MVNOs, we believe the Starlink opportunity could be materially better. In addition, the global reach of Starlink and strength of its broadband offering can create benefits beyond D2D services alone.**

**Most important for SpaceX's valuation remains gaining Starship reusability to enable orbital data centers.** The company now expects second stage reuse to be demonstrated in Q4. Although less critical, we think this reuse will still be important for Starlink broadband and Starlink Mobile constellations beginning in 2027.

**Typical MVNO positioning: a tough business.** MVNOs are generally positioned as lower-cost, value-oriented alternatives to the facilities-based players. This positioning leads to lower pricing, while customer acquisition still requires substantial marketing spend, physical retail footprint and, in many market including the US, aggressive device promos.

Without owner economics, these dynamics tend to produce low-margin and often unattractive standalone businesses. As a result, MVNOs typically target specific customer segments where they can differentiate beyond price alone, such as Consumer Cellular's focus on seniors.

The US CableCos provide an interesting development. Comcast and Charter have built meaningful wireless businesses over the past ~8 years, but only through aggressive pricing, free line offers, device promos, and bundled discounts. Importantly, wireless is not the profit center. It serves primarily to protect and enhance the broadband franchise—at least in the foreseeable future. Viewed on a standalone basis, wireless profitability remains modest, and the economics work because the bundle improves overall CLV. That said, meaningful market share gains are possible. Several MVNOs have captured meaningful market share over time, but it typically comes at the expense of profitability.

**Why Starlink's MVNO economics could be better.** Traditional MVNO economics are constrained by four factors: CAC, device subsidies, lower service pricing, and network access fees paid to MNOs. For Starlink, the first three factors are largely dictated by market structure and competitive dynamics and, therefore, remains inevitable, particularly while the business remains subscale. Network costs, however, may be different.

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The average US wireless subscriber offloads roughly 80% of data consumption to Wi-Fi, leaving around 20% over cellular networks. That remaining traffic is generally what drives wholesale network costs for MVNOs. Cable operators have already demonstrated the ability to push offload rates >85% through Wi-Fi first architectures.

Starlink could potentially achieve even higher levels of offload if a meaningful portion of outdoor mobile usage can be carried directly over its satellite network—"D2D"—or perhaps over a femtocell network in certain areas. Significant engineering challenges remain, including seamless handoffs between terrestrial and satellite networks and the limitations inherent in satellite beamforming and capacity management. But if those challenges are addressed over time, Starlink could materially reduce its reliance on wholesale cellular capacity.

The result is an MVNO model with potentially better economics than a traditional MVNO, particularly when bundled with Starlink's high-margin BB products.

***MVNO structure could support SpaceX's objectives.*** We believe a hybrid MVNO-based model represents the faster and most capital-efficient path toward SpaceX's wireless ambitions. Building a nationwide terrestrial network from scratch would require enormous capital and years of execution, whereas an MVNO model offers a faster path to market.

More importantly, wireless could strengthen Starlink's broader BB ecosystem. By bundling mobile and BB services, SpaceX can expand the reach of its much more profitable O&O connectivity platform while improving customer retention and CLV.

But the larger strategic opportunity may extend beyond wireless economics altogether. In a future where AI becomes ubiquitous and the smartphone in your pocket remains the primary edge device, owning the customer relationship may matter more than maximizing wireless margins. Viewed through that lense, the objective is not necessarily to optimize the economics of a standalone wireless business, whether through an MVNO or a facilities-based network. The objective is to achieve scale, establish distribution, and own the customer relationship as quickly as possible. If that is the goal, an MVNO-based hybrid network may be the most practical path forward.

## BERNSTEIN TICKER TABLE

| Ticker               | Rating | Cur | 11 Sep   | Price  | TTM     | Adjusted EPS |       |        |       | Adjusted P/E (x) |         |       |
|----------------------|--------|-----|----------|--------|---------|--------------|-------|--------|-------|------------------|---------|-------|
|                      |        |     | 2026     |        | Rel.    | Cur          | 2025A | 2026E  | 2027E | 2025A            | 2026E   | 2027E |
|                      |        |     | Closing  | Target | Perf.   |              |       |        |       |                  |         |       |
|                      |        |     | Price    |        |         |              |       |        |       |                  |         |       |
| SPCX (SpaceX)        | O      | USD | 151.21   | 248.00 | NA      | USD          |       | (0.20) | 1.30  |                  | (765.2) | 116.1 |
| T (AT&T)             | O      | USD | 26.06    | 25.00  | (27.9)% | USD          | 2.12  | 2.35   | 2.68  | 12.3             | 11.1    | 9.7   |
| VZ (Verizon)         | M      | USD | 50.61    | 47.00  | (1.6)%  | USD          | 4.68  | 5.00   | 5.20  | 10.8             | 10.1    | 9.7   |
| TMUS (T-Mobile)      | M      | USD | 182.33   | 220.00 | (40.7)% | USD          | 9.72  | 10.68  | 13.24 | 18.8             | 17.1    | 13.8  |
| CMCSA (Comcast)      | M      | USD | 25.20    | 28.00  | (35.7)% | USD          | 4.30  | 3.40   | 3.85  | 5.9              | 7.4     | 6.6   |
| CHTR (Charter)       | M      | USD | 145.77   | 150.00 | (61.0)% | USD          | 36.20 | 38.98  | 40.75 | 4.0              | 3.7     | 3.6   |
| AMT (American Tower) | O      | USD | 177.85   | 214.00 | (25.8)% | USD          | 5.40  | 7.01   | 6.93  | 32.9             | 25.4    | 25.7  |
| CCI (Crown Castle)   | O      | USD | 75.62    | 96.00  | (37.1)% | USD          | 1.02  | 2.01   | 3.22  | 74.4             | 37.7    | 23.5  |
| SBAC (SBA Comm)      | M      | USD | 185.47   | 219.00 | (23.2)% | USD          | 11.76 | 8.81   | 9.57  | 15.8             | 21.0    | 19.4  |
| SPX                  |        |     | 7,656.98 |        |         |              |       |        |       |                  |         |       |

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

## INVESTMENT IMPLICATIONS

We rate SpaceX Outperform, with a target of \$248. The case for a multi-trillion dollar valuation continues to be first about delivering on the orbital data center opportunity. Still, we forecast the Connectivity business to generate more than \$100bn in EBITDA by 2031, with more upside possible if the D2D business is successful - which we believe will require a strong MVNO solution. The case for SpaceX was described in our initiation report ([Space Exploration Technologies Corporation: Doing Time in the Universal Mind - Initiate at Outperform; Target Price \\$239](#)) and updated after Q2 reporting ([SpaceX: Solid Q2 with no issues; Projects more upside with higher compute pricing; Starship described on track - Outperform](#)).

We maintain our Outperform rating for T (PT: \$25), and Market-Perform rating for VZ (PT: \$47), TMUS (PT: \$220), CMCSA (PT: \$28), and CHTR (PT: \$150) - all unchanged.

We maintain our Outperform ratings for AMT (PT: \$214) and CCI (PT: \$96) and our Market-Perform rating for SBAC (PT: \$219).

## DETAILS

### THE GLOBAL WIRELESS MARKET IS MOSTLY SATURATED

In developed markets, network differentiation has become increasingly difficult. Every operator claims to be “#1” on some metric, making meaningful performance advantages harder for consumers to discern. As a result, carriers increasingly compete through brand positioning, customer segmentation, pricing, and bundling rather than network superiority alone.

#### US Wireless Market is Saturated

From the rise of the internet in the 1990s to the mobile era of the late 2000s, the world underwent a profound digital transformation. Smartphones became ubiquitous, and internet connectivity is now an essential part of daily life. Today, nearly every adult in the U.S. owns a smartphone, with many maintaining separate lines for personal and work use. The introduction of connected devices such as tablets and smartwatches has further increased the prevalence of multiple wireless connections per person.

Smartphone ownership has also become increasingly common among teenagers, while children are obtaining mobile service at younger ages. Together, these trends have pushed the U.S. wireless market toward saturation ([Exhibit 1](#)). Current market penetration has already reached approximately 90%.

Despite operating in a saturated market, the total number of wireless connections has continued to grow at roughly 2% annually. We expect this trend to persist, driven by three fundamental factors: population growth, increases in penetration rates, and greater adoption of multiple connected devices. Higher penetration is primarily supported by rising smartphone adoption among younger age cohorts, while multi-device adoption reflects both the growing use of separate work and personal lines and the increasing penetration of connected devices that require wireless connectivity.

That said, the market's long-term growth potential remains limited. As a result, the emergence of a new large-scale share gainer, such as a Starlink MVNO service, may face challenges. The cable operators provide a useful precedent, as they successfully followed this strategy and gained market share over time.

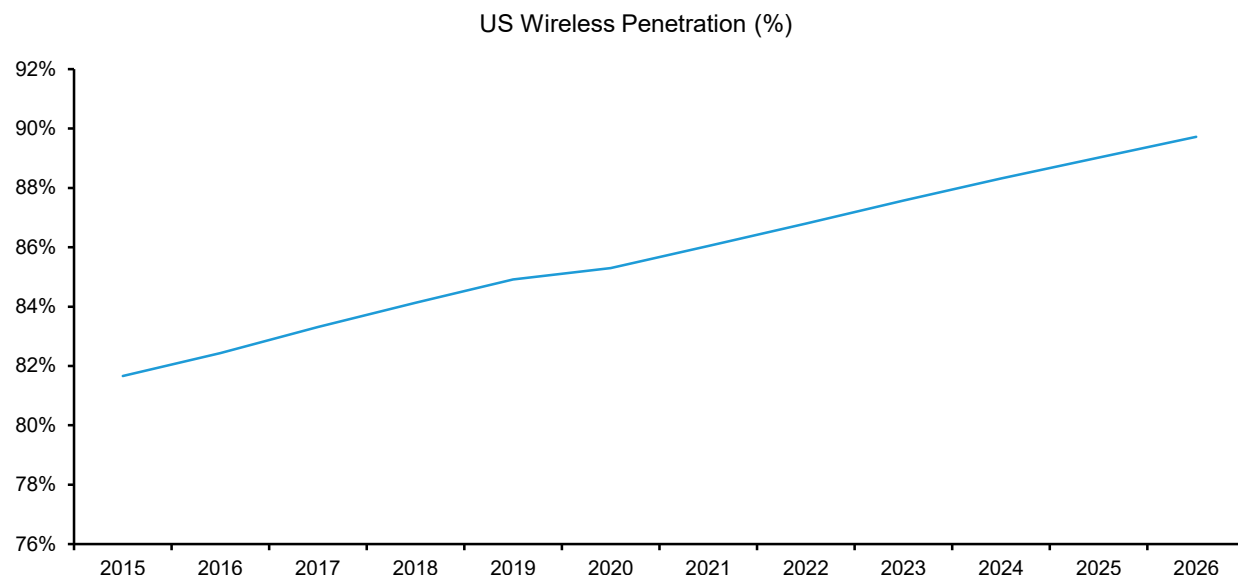
### INTERNATIONAL WIRELESS MARKETS ARE LARGELY SATURATED, BUT POCKETS OF UNDERPENETRATION REMAIN

Most international wireless markets also exhibit signs of saturation. As a point of reference, the ratio of total mobile subscriptions to population for several key markets is: the U.S. (133%), Germany (166%), France (144%), Japan (170%), China (125%), and Chile (164%).

However, several large geographic markets, including Canada (106%), Australia (102%), India (82%), and Brazil (102%), appear relatively underpenetrated compared with their peers. These markets may present an opportunity for a provider that can leverage satellite coverage as a complement to a terrestrial MVNO agreement, such as Starlink ([Exhibit 2](#)).

**EXHIBIT 1: The US wireless market is already saturated at ~90% penetration. The opportunity for a Starlink MVNO service is in taking share from incumbents**

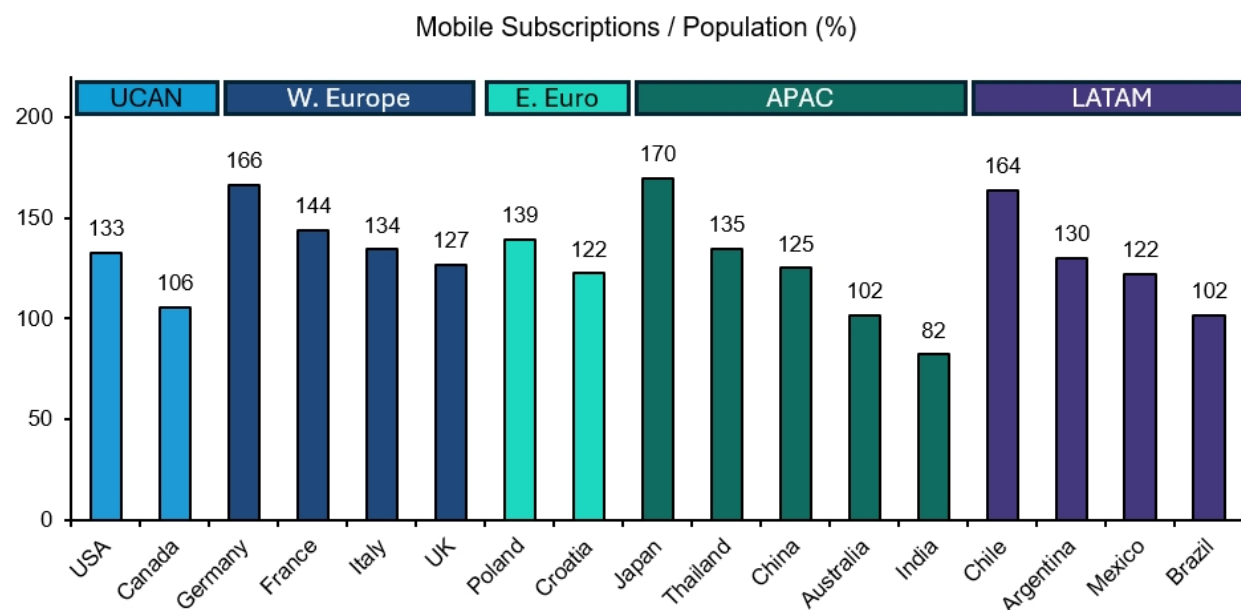
Adjusts for individuals with more than 1 line.



Source: SNL Kagan, Bernstein estimates and analysis

**EXHIBIT 2: Most developed and developing markets are saturated or nearing saturation**

Data for African markets not available. Does not adjust for individuals with more than 1 line.



Source: SNL Kagan, Bernstein estimates and analysis

## THE MVNO ECOSYSTEM

MVNO (Mobile Virtual Network Operator) agreements, in which a Mobile Network Operator (MNO) sells access to its network at wholesale rates, are common across global wireless markets. MVNOs typically market wireless services under their own brands, own the customer relationship, and compensate MNOs based on the network usage generated by their subscribers.

These arrangements are generally attractive to MNOs when the MVNO can acquire customer segments that the carrier would not otherwise be able to reach efficiently.

In the U.S., all three major wireless carriers, Verizon, AT&T, and T-Mobile, participate in MVNO partnerships. Notable independent MVNO relationships include Xfinity Mobile, Spectrum Mobile, and Cox Mobile on Verizon's network; Consumer Cellular, Boost Mobile, and H2O Wireless on AT&T's network; and Optimum Mobile, and Astound Mobile on T-Mobile's network. The model is particularly important for Verizon, which maintains the industry's two largest MVNO agreements with Comcast (Xfinity Mobile) and Charter Communications (Spectrum Mobile) ([Exhibit 3](#)).

For broadband providers such as Comcast, Charter, Optimum, and Cox, entering into MVNO agreements is often attractive because customer acquisition costs can be significantly lower than those of a traditional wireless carrier. By cross-selling mobile services to existing broadband subscribers, these companies can leverage established customer relationships, brand awareness, and billing platforms, reducing both marketing expenses and customer churn.

### EXHIBIT 3: Examples of US MVNOs: Verizon has the largest individual agreements through CMCSA (Xfinity) and CHTR (Spectrum)

|                                      | Verizon                                                                                                             | AT&T                                                                                                                  | T-Mobile                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Major Independent MVNO Relationships | <ul style="list-style-type: none"> <li>• Xfinity Mobile</li> <li>• Spectrum Mobile</li> <li>• Cox Mobile</li> </ul> | <ul style="list-style-type: none"> <li>• Consumer Cellular</li> <li>• Boost Mobile</li> <li>• H2O Wireless</li> </ul> | <ul style="list-style-type: none"> <li>• Optimum Mobile</li> <li>• Astound Mobile</li> </ul> |

Source: Company websites, Company filings, Bernstein analysis

## CAN MVNOS SCALE?

### MVNOs can take meaningful market share

Despite U.S. wireless subscriber growth averaging only about 2% annually, Charter and Comcast have grown their mobile subscriber bases at an approximately 60% CAGR over the past decade through their MVNO strategies ([Exhibit 4](#)). Through their MVNO agreements with Verizon, Charter and Comcast have increased their combined share of the postpaid phone market from zero to approximately 7% in less than a decade. During the same period, Verizon's market share declined from roughly 40% to the low-30% range ([Exhibit 5](#)).

The experience is not unique to the U.S.. In international markets such as the UK and Italy, MVNOs have collectively captured market shares in the high teens. A Starlink MVNO service could potentially expand wireless penetration in certain underserved markets where satellite connectivity complements existing terrestrial networks ([Exhibit 6](#)).

The success of the cable operators' wireless businesses, coupled with the corresponding loss of market share by their host network partner, Verizon, raises an important question: are large MVNO agreements ultimately beneficial for MNOs, or do they primarily cannibalize the carriers' natural addressable customer base? This question becomes particularly relevant if MNOs have to consider a future partnership with Starlink.

### MVNOs often rely on aggressive offers to gain share

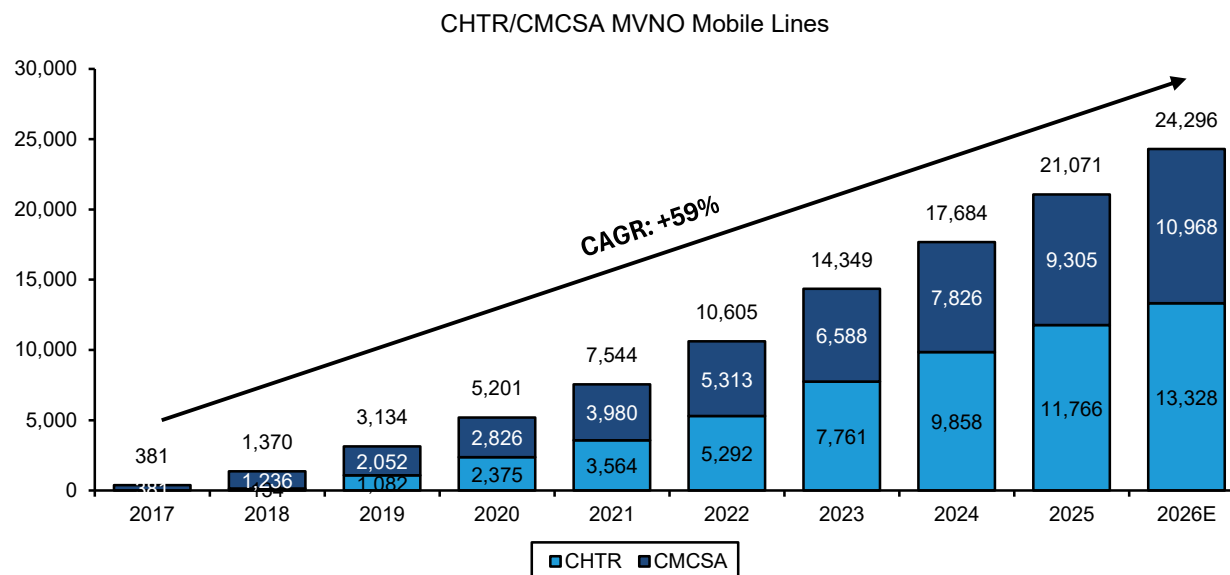
To encourage customers to switch from well-established MNO brands, MVNOs frequently rely on highly competitive promotional offers. These promotions typically include a combination of free service for a period of time, lower monthly rates than the host MNO, and attractive device subsidies.

For example, Xfinity Mobile's value-oriented offering generally provides lower device subsidies than those offered by the major MNOs. However, the service is free during the first year and remains materially cheaper thereafter. As a result, customers can

save approximately \$45 to \$60 per month during the first year of service ([Exhibit 7](#)).

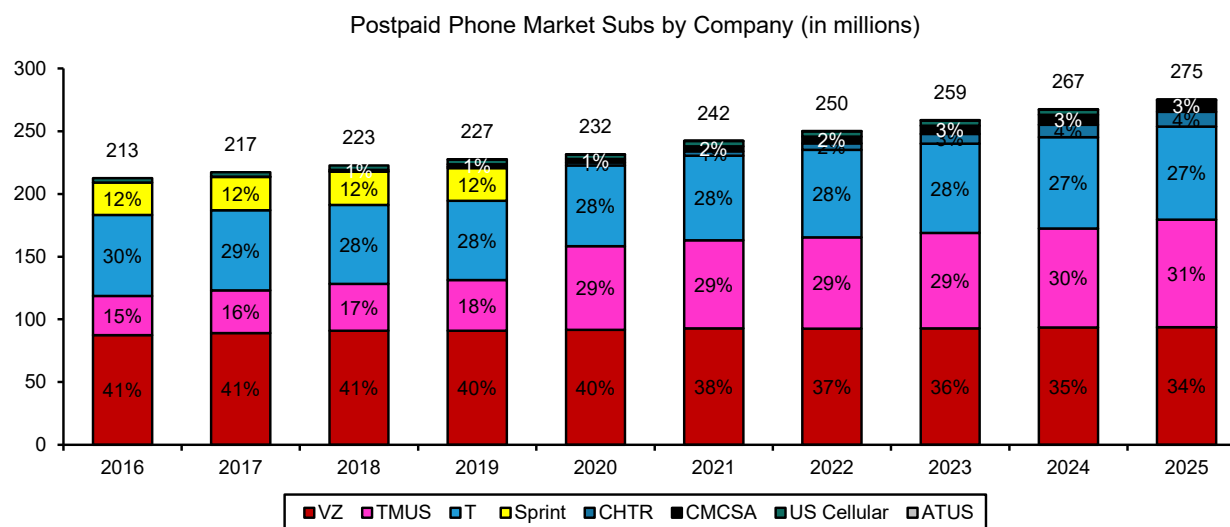
Under Xfinity Mobile's premium plan, customers pay a discounted monthly fee during the first year, followed by a relatively low ongoing rate. In addition, device subsidies are broadly comparable to those offered by the national carriers. Under this plan, customers can save approximately \$95 to \$105 per month during the first year ([Exhibit 8](#)).

#### EXHIBIT 4: CableMVNOs have grown rapidly in the US market

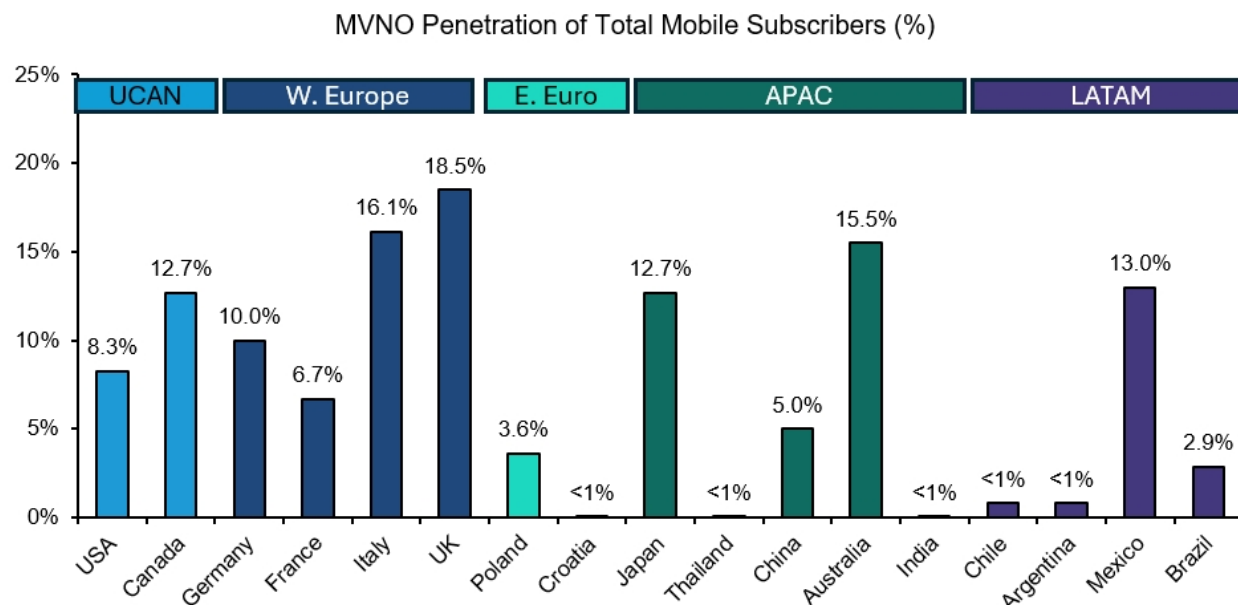


Source: Company Filings, Bloomberg, Bernstein estimates and analysis

#### EXHIBIT 5: CHTR and CMCSA went from 0% to ~7% of the postpaid phone market subs in less than ten years through MVNO agreements with Verizon



Source: Company Filings, Bloomberg, Bernstein estimates and analysis

EXHIBIT 6: **Some markets have meaningful MVNO market shares**

Source: SNL Kagan, Bernstein estimates and analysis

EXHIBIT 7: **Xfinity's MVNO model offers less device subsidies with its value plan than the MNOs, but the service is free in year 1, and considerably cheaper afterwards**

|                                                           |                                                                                                                                            |                                                                                                                                       |                                                                                                            |                                                                                              |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                                                           | New Phone<br>iPhone 17                                                                                                                     | Trade-in Phone<br>iPhone 16                                                                                                           | Xfinity Mobile Plan<br>Mobile Select                                                                       | Xfinity's value plan                                                                         |
|                                                           | <b>Xfinity</b>                                                                                                                             | <b>AT&amp;T</b>                                                                                                                       | <b>T-Mobile</b>                                                                                            | <b>Verizon</b>                                                                               |
| Less device subsidy than competitors                      | PHONE<br><b>iPhone 17</b><br>\$14.06/mo for 36 mos<br>Includes \$324.00 trade-in credit                                                    | PHONE<br><b>iPhone 17</b><br>\$9.16/mo for 36 mos<br>Includes \$500.00 trade-in credit                                                | PHONE<br><b>iPhone 17</b><br>\$2.77/mo for 36 mos<br>Includes \$730.00 trade-in credit                     | PHONE<br><b>iPhone 17</b><br>\$6.38/mo for 36 mos<br>Includes \$600.00 trade-in credit       |
| Service is free for 1 year, cheaper than comps afterwards | MOBILE PLAN<br><b>Mobile Select</b><br>\$99-\$0/mo/1 line<br>Includes \$30/mo credit for 12 months                                         | MOBILE PLAN<br><b>Value 2.0</b><br>\$50/mo/1 line                                                                                     | MOBILE PLAN<br><b>Essentials 2.0</b><br>\$60/mo/1 line                                                     | MOBILE PLAN<br><b>Unlimited Welcome</b><br>\$65/mo/1 line                                    |
|                                                           | Plan Features ^<br>• Data: Unlimited high-speed (may slow after 50GB)<br>• Hotspot: 15GB high-speed data, then unlimited at reduced speeds | Plan Features ^<br>• Data: Unlimited (may slow during congestion)<br>• Hotspot: 3GB high-speed data, then unlimited at reduced speeds | Plan Features ^<br>• Data: Unlimited (reduced speeds after 50GB)<br>• Hotspot: Unlimited at reduced speeds | Plan Features ^<br>• Data: Unlimited (may slow during congestion)<br>• Hotspot: Not included |
|                                                           | TOTAL FOR PHONE + MOBILE PLAN<br><b>\$14.06 /mo</b><br>Taxes and fees are not included                                                     | TOTAL FOR PHONE + MOBILE PLAN<br><b>\$59.16 /mo</b><br>Taxes and fees are not included                                                | TOTAL FOR PHONE + MOBILE PLAN<br><b>\$62.77 /mo</b><br>Taxes and fees are not included                     | TOTAL FOR PHONE + MOBILE PLAN<br><b>\$71.38 /mo</b><br>Taxes and fees are not included       |
|                                                           | <b>Shop This Deal</b><br>Xfinity Internet Required                                                                                         | Xfinity saves you up to<br><b>\$45 /mo</b>                                                                                            | Xfinity saves you up to<br><b>\$49 /mo</b>                                                                 | Xfinity saves you up to<br><b>\$57 /mo</b>                                                   |
|                                                           |                                                                                                                                            | Annual savings decline by \$30 after the 1 <sup>st</sup> year                                                                         |                                                                                                            |                                                                                              |

Savings comparison based on 1 line of Xfinity Mobile Select vs 1 line of T-Mobile Essentials, Verizon Unlimited Welcome, and AT&amp;T Extra 2.0 (as of 03/24/26). Taxes &amp; fees extra.

Source: Company website, Bernstein analysis

**EXHIBIT 8: In its premium MVNO plan, Xfinity's customers pay a discounted fee on year 1 with a cheaper run rate fee afterwards. Additionally, device subsidies are on par with MNOs**

|                              | Xfinity                                                                                                                                               | AT&T                                                                                                                                                   | Verizon                                                                                                                                                | T-Mobile                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHONE                        | iPhone 17<br>\$0.00/mo for 36 mos<br>Includes \$829.99 trade-in credit<br>Device protection: Included                                                 | iPhone 17<br>\$3.61/mo for 36 mos<br>Includes \$700.00 trade-in credit<br>Device protection: \$19/mo                                                   | iPhone 17<br>\$0.00/mo for 36 mos<br>Includes \$830.00 trade-in credit<br>Device protection: \$19/mo                                                   | iPhone 17<br>\$0.00/mo for 36 mos<br>Includes \$830.00 trade-in credit<br>Device protection: \$19/mo                                                   |
| MOBILE PLAN                  | Mobile Plus<br>\$45-\$15/mo/1 line<br>Includes \$30/mo credit for 12 months                                                                           | Premium 2.0<br>\$109/mo/1 line                                                                                                                         | Unlimited Ultimate<br>\$114/mo/1 line                                                                                                                  | Experience Beyond 2.0<br>\$119/mo/1 line                                                                                                               |
| Plan Features                | <ul style="list-style-type: none"> <li>Data: Unlimited high-speed</li> <li>Hotspot: 50GB high-speed data, then unlimited at reduced speeds</li> </ul> | <ul style="list-style-type: none"> <li>Data: Unlimited high-speed</li> <li>Hotspot: 100GB high-speed data, then unlimited at reduced speeds</li> </ul> | <ul style="list-style-type: none"> <li>Data: Unlimited high-speed</li> <li>Hotspot: 200GB high-speed data, then unlimited at reduced speeds</li> </ul> | <ul style="list-style-type: none"> <li>Data: Unlimited high-speed</li> <li>Hotspot: 250GB high-speed data, then unlimited at reduced speeds</li> </ul> |
| TOTAL WITH DEVICE PROTECTION | <b>\$15.00</b> /mo<br>Taxes and fees are not included                                                                                                 | <b>\$112.61</b> /mo<br>Taxes and fees are not included                                                                                                 | <b>\$114.00</b> /mo<br>Taxes and fees are not included                                                                                                 | <b>\$119.00</b> /mo<br>Taxes and fees are not included                                                                                                 |
|                              | Shop This Deal<br>Xfinity Internet Required                                                                                                           | Xfinity saves you up to <b>\$98</b> /mo                                                                                                                | Xfinity saves you up to <b>\$99</b> /mo                                                                                                                | Xfinity saves you up to <b>\$104</b> /mo                                                                                                               |

Annual savings decline by \$30 after the 1<sup>st</sup> year

Source: Company website, Bernstein analysis

## STANDARD MVNO ECONOMICS VS STARLINK MVNO ECONOMICS

### Lack of network ownership usually results in modest profitability

One of the most important variables in MVNO economics is the wholesale fee that the MNO charges the MVNO for access to its network. In a simplified framework, the MNO and MVNO agree on a rate per gigabyte (GB) of data consumed. The MVNO's total network expense is then determined by multiplying this rate by the volume of data usage generated by its subscribers. All else being equal, an MVNO's cost structure improves when it can offload traffic away from the host MNO's network. For example, when a subscriber uses home Wi-Fi rather than the terrestrial mobile network, the MVNO avoids incurring wholesale network costs on that usage ([Exhibit 9](#)).

To reduce wholesale expenses, MVNOs seek to maximize traffic offloading. In the U.S., approximately 80-90% of handset data traffic is already offloaded to Wi-Fi as a result of consumer usage patterns, since the most data-intensive activities typically occur at home, in the office, or in other locations with Wi-Fi access. Cable MVNOs such as Comcast and Charter have been able to achieve approximately 87-90% traffic offloading through a combination of extensive Wi-Fi hotspot networks and owned CBRS spectrum network, representing roughly 2-7 percentage points of incremental offloading relative to traditional MNOs. A potential Starlink MVNO offering could leverage its satellite constellation and femtocells to offload traffic in away-from-home use cases. Unlike cable operators' CBRS networks, satellite coverage is largely ubiquitous, potentially enabling even higher levels of traffic offloading ([Exhibit 10](#)).

Because MVNOs typically rely on aggressive promotional activity while lacking full network owner economics, their margins are generally lower than those of MNOs. The most recent disclosed figure was Charter's wireless EBITDA margin of approximately -11% in FY22, when the business had significantly fewer subscribers than it does today. Ultimately, MVNO profitability depends on several factors, including the degree of traffic offloading, the level of wholesale rates, and operating scale. In addition, SG&A efficiency generally improves as subscriber bases grow. As the two largest MVNOs in the U.S., Comcast and Charter have both achieved positive EBITDA in their mobile businesses as scale has increased ([Exhibit 11](#)).

Starlink's ability to offload MVNO traffic onto its own satellite network in away-from-home use cases could produce meaningful cost savings under an MVNO model. For example, increasing traffic offloading from approximately 88%, which is similar to current cable MVNO levels, to 91% would reduce wholesale MVNO fees by roughly 25%, all else being equal ([Exhibit 12](#)).

Relatively small increases in offloading can therefore generate substantial savings. These savings could translate into higher MVNO margins for Starlink, support higher wholesale rates charged by MNO partners, and/or enable lower prices for end customers ([Exhibit 13](#)).

### **An MVNO Strategy May Make Sense for Starlink if It Expands the Broadband Opportunity**

The cash flow profile of a Starlink MVNO customer could vary significantly depending on the go-to-market strategy and customer adoption patterns. Device subsidies, promotional credits, and handset installment plans typically represent a large portion of customer acquisition costs. For example, assume Starlink acquires a new mobile account with two lines, and both subscribers purchase a new \$1,100 smartphone. If Starlink offers a trade-in credit equal to approximately 75% of the new device value (\$825 per device), while the traded-in devices can only be resold for roughly 60% of the new device value (\$660), the transaction would create an immediate initial customer acquisition outflow of \$880  $((\$1,100 - \$660) \times 2)$  and a device-related loss of approximately \$330  $((\$825 - \$660) \times 2)$ . In addition, financing the new devices over a 36-month installment period would create an additional working-capital burden and time-value-of-money cost.

Assuming Starlink prices mobile service at approximately \$35 per line per month, similar to Comcast and Charter, and achieves wireless EBITDA margins of roughly 25% due to greater traffic offloading than cable MVNOs, the company could generate approximately \$650 of cumulative profit over the first three years of the customer relationship after accounting for pricing increases, account growth, and margin expansion.

The overall economics of an MVNO strategy also depend on broadband bundling dynamics. Some mobility customers may not subscribe to Starlink broadband services, while others may bundle broadband and wireless in exchange for promotional discounts. For bundled customers, two outcomes are possible. First, the customer may already be a Starlink broadband subscriber, in which case the primary benefit is likely lower broadband churn, partially offset by the bundled discount. Second, the customer may be entirely new to the broadband business, creating incremental service revenue that otherwise would not have existed (potentially expanding the TAM). The proportion of customers that fall into each category is a critical determinant of the overall economic value of the bundle (and whether the net bundle economics are either positive or negative).

Importantly, the customer relationship does not necessarily end once the initial 36-month handset financing cycle is complete. The upgrade and replacement cycle may repeat, creating another multiyear stream of customer cash flows and acquisition costs ([Exhibit 14](#)).

As discussed above, the internal rate of return (IRR) of a Starlink MVNO strategy depends on several variables. Two of the most important are: (1) the trade-in credit offered as a percentage of new device value and (2) among customers who bundle broadband and wireless services, the percentage that are new broadband subscribers rather than existing Starlink broadband customers. We assume a 40% broadband bundle rate, which is broadly consistent with AT&T's fiber and wireless bundling penetration.

Under the assumptions presented in [Exhibit 15](#), the IRR approaches 0% when the trade-in value offered is equal to the full value of the new device, effectively eliminating future customer device payments. If trade-in credits exceed the value of the new devices through additional incentives, such as prepaid cards or promotional credits, the IRR becomes negative. Conversely, when trade-in values are lower than the price of new devices, the economics improve and the IRR turns positive. A higher proportion of new broadband customers within bundled accounts can further enhance returns. Under the right combination of customer acquisition costs, traffic offloading benefits, and broadband attachment rates, a Starlink MVNO strategy could generate returns above its cost of capital.

### **Owning the customer relationship may matter more than maximizing wireless margins**

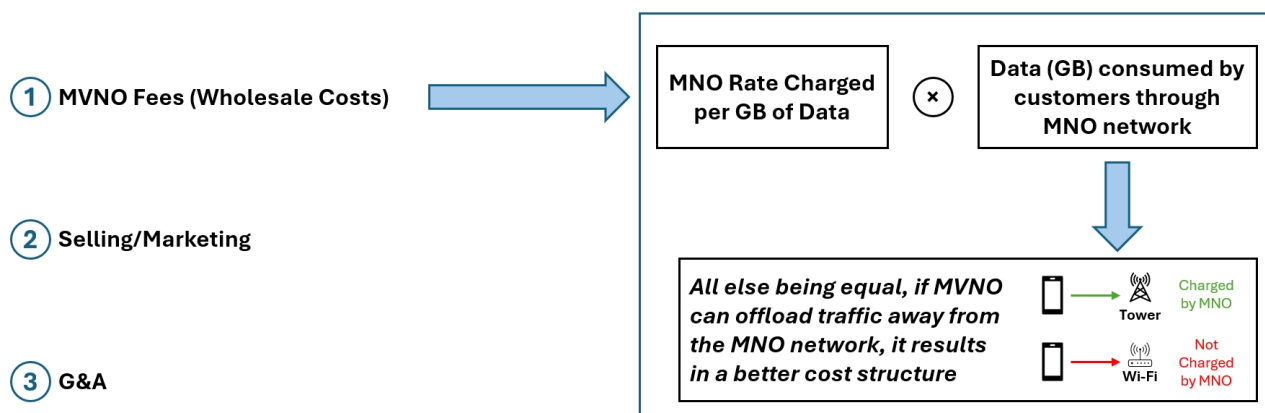
There are at least three potential objectives for SpaceX in developing a wireless business. The first is the growth opportunity that a successful standalone wireless service could provide at scale. Achieving this would require Starlink to capture a meaningful share of the wireless market. Starlink's MVNO economics could be materially stronger than those of a typical MVNO due to D2D offloading, potentially supporting this objective.

The second objective is to expand Starlink's broadband addressable market. In many markets, connectivity is already, or is increasingly becoming, a converged product, with broadband and mobility services sold together. Offering broadband-only services can limit the total addressable market if a converged offering is unavailable. An MVNO structure would enable Starlink to bundle wireless and broadband services, increasing both customer penetration and retention in broadband, similar to the strategy employed by U.S. cable operators today.

However, the larger strategic opportunity may extend beyond wireless economics altogether. In a future where AI becomes ubiquitous and the smartphone remains the primary edge device, owning the customer relationship may matter more than maximizing wireless margins. Viewed through this lens, the objective is not necessarily to optimize the economics of a standalone wireless business, whether through an MVNO arrangement or a facilities-based network. Rather, the goal may be to achieve scale, establish distribution, and build direct customer relationships as quickly as possible. If that is the objective, an MVNO-based hybrid network could represent the most practical path forward ([Exhibit 16](#)).

#### EXHIBIT 9: MVNO's profitability is dependent on its ability to offload traffic away from the MNO network

##### Elements of MVNO's Cost to Serve

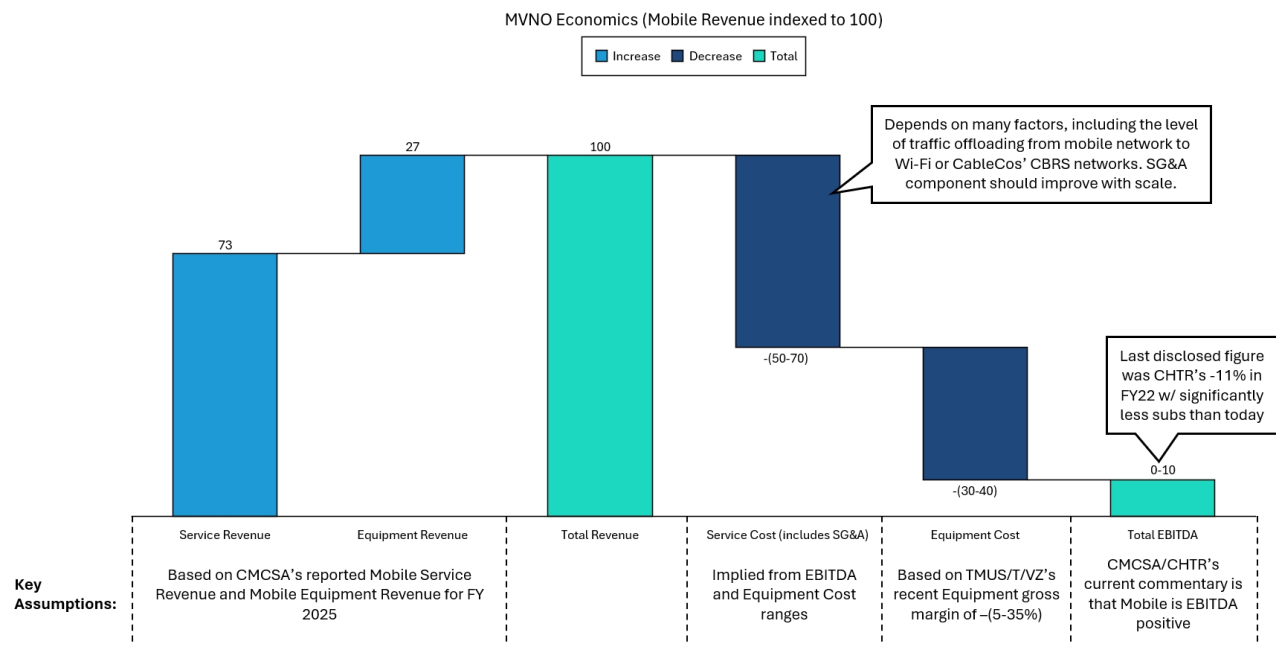


Source: Bernstein estimates and analysis

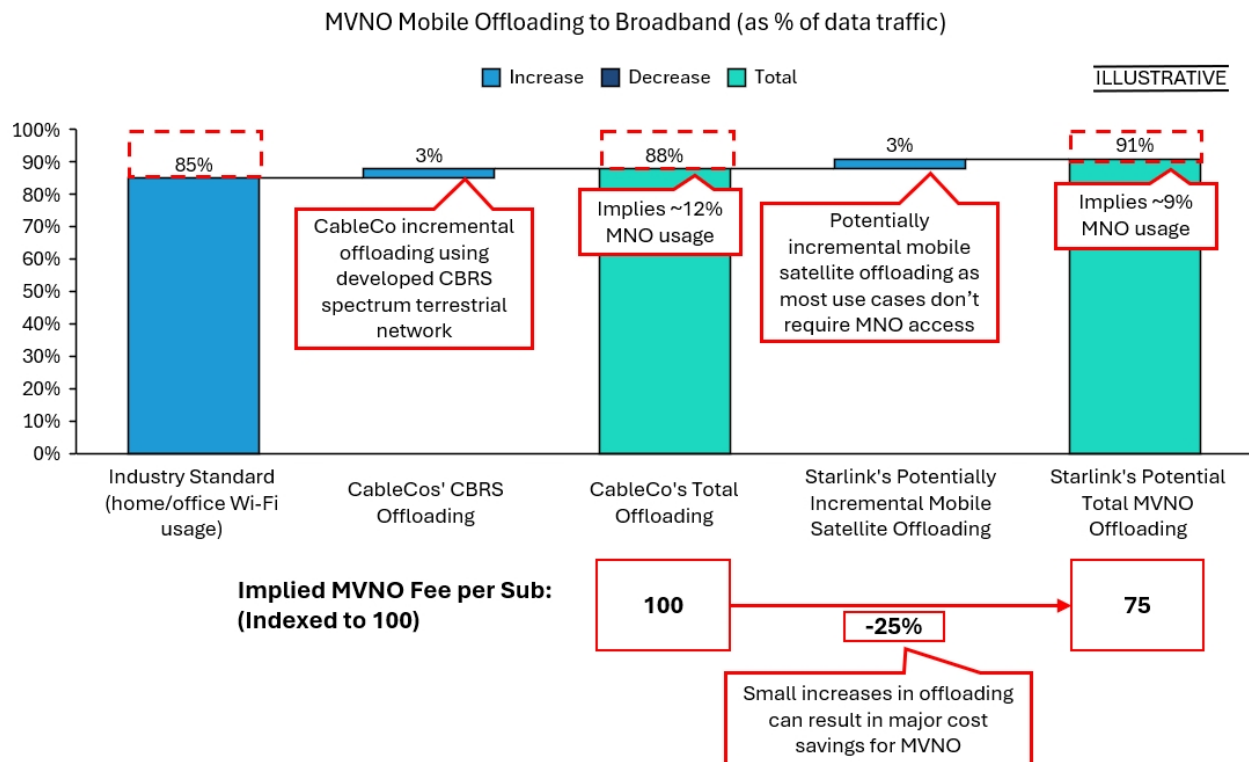
#### EXHIBIT 10: A potential Starlink MVNO model could have an advantage over CableCo's in that it may be able to offload significant traffic to its satellite constellation and - perhaps - femtocells

|                                                                                                                                           | Potential Differences in MVNO Offloading                                                            |                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                                                                                                                           | Cable MVNOs                                                                                         | Potential Starlink MVNO                                                                                        |
| <b>Normal US Handset Offloading to Wi-Fi</b>                                                                                              | ~80-90% of handset data traffic is offloaded to Wi-Fi due to consumption patterns in the US         |                                                                                                                |
| <b>Wi-Fi Hotspots &amp; CBRS Spectrum Terrestrial Network</b>                                                                             | CHTR & CMCSA achieve ~87-90% offloading to their extensive Wi-Fi and CBRS networks (+2-7% vs. MNOs) | No owned terrestrial network                                                                                   |
| <b>Starlink Mobile Satellite Constellation/Femtocell</b>                                                                                  | No owned satellite network                                                                          | Starlink could potentially offload traffic to its satellite network or femtocells for away-from-home use cases |
| Unlike CableCo's CBRS network, the satellite constellation & femtocell combination is ubiquitous, which could result in higher offloading |                                                                                                     |                                                                                                                |

Source: OpenSignal, Company filings, Bernstein estimates and analysis

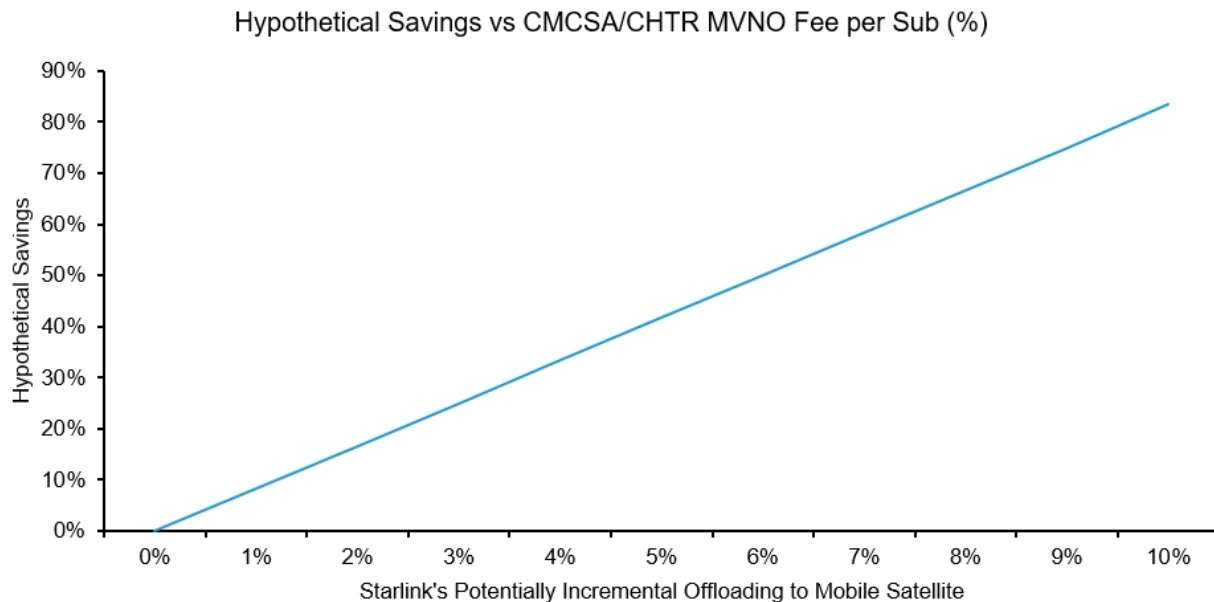
**EXHIBIT 11: CMCSA and CHTR have achieved positive EBITDA on their Mobile businesses as the largest MVNOs in the US**


Source: Company filings, Bernstein estimates and analysis

**EXHIBIT 12: Starlink's capacity to offload MVNO traffic to its own mobile satellite network in away-from-home use cases could result in major cost savings in an MVNO model**


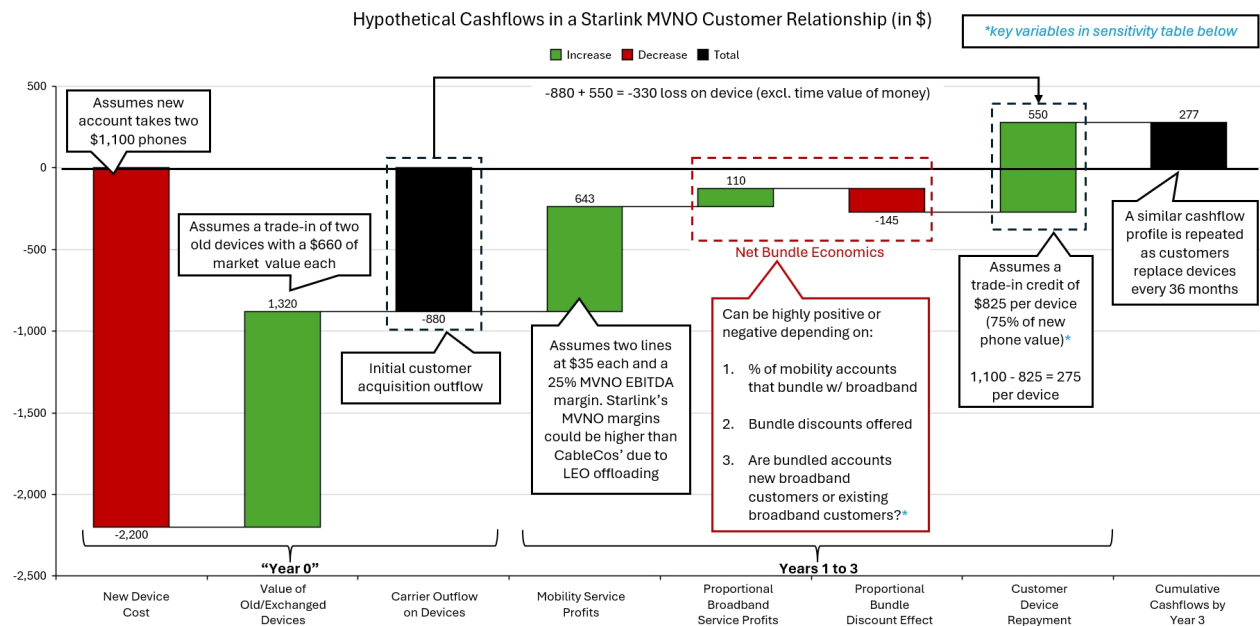
Source: OpenSignal, Company filings, Bernstein estimates and analysis

**EXHIBIT 13: Small offloading increments can result in major cost savings for Starlink. This can result in better MVNO margins for Starlink, higher fees per bit charged by MNOs, and/or lower prices for Starlink Mobile customers**

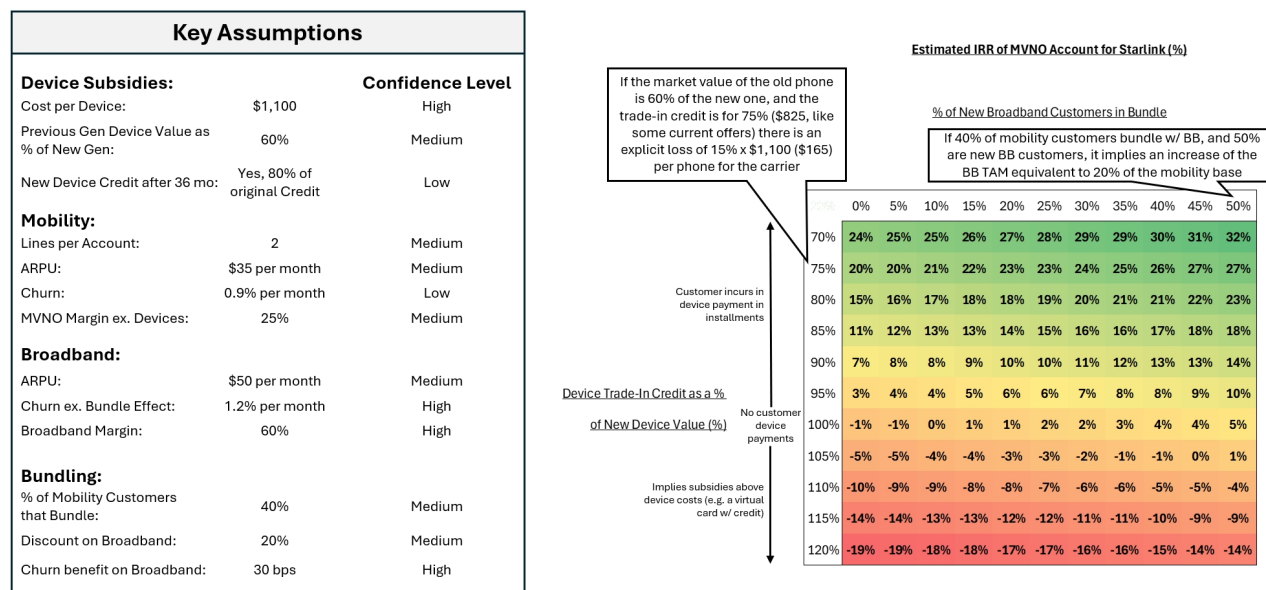


Source: Company filings, Bernstein estimates and analysis

**EXHIBIT 14: The cashflow profile of a Starlink MVNO customer relationship may look very different depending on strategy and market reception. One key variable is by what degree do bundles increase the TAM for broadband customers**



Source: Company websites, Bernstein estimates and analysis

EXHIBIT 15: **The Starlink wireless business could expand its broadband TAM and produce better returns****Analysis of Estimated IRR of MVNO Account for Starlink**

Source: Company websites, Company filings, Bernstein estimates and analysis

EXHIBIT 16: **The more important strategic opportunity for a Starlink MVNO may extend beyond wireless economics altogether. If AI becomes ubiquitous, owning the customer relationship may matter more than maximizing wireless margins**

| Potential objectives of Wireless for SpaceX                                                                                                                                                                                                                              | Does MVNO structure support?                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wireless growth</b> <ul style="list-style-type: none"> <li>Standalone merits of wireless at scale can be attractive</li> <li>Would need to believe Starlink can capture a meaningful share of the market</li> </ul>                                                   | <input checked="" type="checkbox"/> <ul style="list-style-type: none"> <li>Standalone MVNO economics is unattractive and challenging to scale</li> <li>However, <b>economics could be materially better</b> than typical MVNOs with D2D offloading</li> </ul> |
| <b>Expand Starlink's BB addressable market</b> <ul style="list-style-type: none"> <li>Many markets, including the US, is already or increasingly converged</li> <li>BB-only play limits TAM</li> </ul>                                                                   | <input checked="" type="checkbox"/> <ul style="list-style-type: none"> <li><b>Wireless-BB bundled offers can increase Starlink BB penetration and retention</b> – works even with low MVNO margins (like US CableCos)</li> </ul>                              |
| <b>Own the customer for future AI-play</b> <ul style="list-style-type: none"> <li>If/when AI is ubiquitous, owning the ultimate edge device can be a strategic advantage to monetize</li> <li>SpaceX's AI can be an embedded service for Starlink Mobile subs</li> </ul> | <input checked="" type="checkbox"/> <ul style="list-style-type: none"> <li><b>Starlink would own the customer relationship</b> (not MNO providing network as a service to Starlink)</li> </ul>                                                                |

Can Starlink-branded MVNO bundled with Starlink BB achieve meaningful scale to achieve its objectives?

Possibly more than traditional MVNOs

Source: Bernstein analysis

## APPENDIX - FINANCIAL FORECASTS

## EXHIBIT 17: SpaceX Annual Income Statement

| SpaceX Annual Income Statement (\$m)       |                  |                   |                   |                   |                   |                   |                   |                   |
|--------------------------------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Fiscal Year Ending December 31             | 2024             | 2025              | 2026E             | 2027E             | 2028E             | 2029E             | 2030E             | 2031E             |
| Space                                      | 3,796            | 4,086             | 4,794             | 7,560             | 7,513             | 7,224             | 7,930             | 9,167             |
| Connectivity                               | 7,599            | 11,387            | 17,037            | 27,727            | 53,120            | 94,679            | 140,998           | 205,908           |
| AI                                         | 2,620            | 3,201             | 24,571            | 115,088           | 195,835           | 229,164           | 305,581           | 439,753           |
| <b>Total Revenues</b>                      | <b>\$ 14,015</b> | <b>\$ 18,674</b>  | <b>\$ 46,402</b>  | <b>\$ 150,375</b> | <b>\$ 256,469</b> | <b>\$ 331,067</b> | <b>\$ 454,509</b> | <b>\$ 654,827</b> |
| Cost of revenue                            | (7,996)          | (9,451)           | (17,167)          | (46,740)          | (80,978)          | (101,693)         | (124,050)         | (178,953)         |
| Research & Development                     | (3,464)          | (8,643)           | (23,201)          | (67,204)          | (49,285)          | (61,761)          | (77,515)          | (110,456)         |
| Selling, General, and administrative       | (1,813)          | (2,644)           | (4,309)           | (8,742)           | (13,599)          | (16,750)          | (22,213)          | (29,195)          |
| Restructuring charges                      | (213)            | (487)             | 9                 | -                 | -                 | -                 | -                 | -                 |
| Impairment                                 | (63)             | (38)              | -                 | -                 | -                 | -                 | -                 | -                 |
| <b>Total cost and expenses</b>             | <b>-13,549</b>   | <b>-21,263</b>    | <b>-44,669</b>    | <b>-122,687</b>   | <b>-143,862</b>   | <b>-180,204</b>   | <b>-223,778</b>   | <b>-318,604</b>   |
| Space                                      | 21               | (657)             | (2,047)           | 1,262             | 896               | 236               | 503               | 874               |
| Connectivity                               | 2,006            | 4,423             | 6,082             | 12,006            | 24,626            | 44,437            | 67,304            | 102,316           |
| AI                                         | (1,561)          | (6,355)           | (2,302)           | 14,419            | 87,084            | 106,190           | 162,924           | 233,033           |
| <b>Operating income</b>                    | <b>\$ 466</b>    | <b>\$ (2,589)</b> | <b>\$ 1,734</b>   | <b>\$ 27,688</b>  | <b>\$ 112,607</b> | <b>\$ 150,863</b> | <b>\$ 230,731</b> | <b>\$ 336,223</b> |
| Interest income (expense)                  | (1,209)          | (1,453)           | (2,129)           | (11,511)          | (16,320)          | (22,010)          | (20,989)          | (20,796)          |
| Other income (expense), net                | 985              | -177              | -1962             | 0                 | 0                 | 0                 | 0                 | 0                 |
| <b>Income (loss) before income taxes</b>   | <b>\$ 242</b>    | <b>\$ (4,219)</b> | <b>\$ (2,357)</b> | <b>\$ 16,177</b>  | <b>\$ 96,287</b>  | <b>\$ 128,853</b> | <b>\$ 209,741</b> | <b>\$ 315,427</b> |
| Provisions for (benefit from) income taxes | 549              | (718)             | (82)              | (99)              | 14,202            | (2,414)           | (4,828)           | (39,563)          |
| <b>Net income</b>                          | <b>\$ 791</b>    | <b>\$ (4,937)</b> | <b>\$ (2,439)</b> | <b>\$ 16,078</b>  | <b>\$ 110,489</b> | <b>\$ 126,439</b> | <b>\$ 204,913</b> | <b>\$ 275,864</b> |
| <b>Shares outstanding (mn)</b>             |                  |                   |                   |                   |                   |                   |                   |                   |
| Basic                                      |                  |                   | 11,789            | 11,789            | 11,789            | 11,789            | 11,789            | 11,789            |
| Diluted                                    |                  |                   | 12,344            | 12,344            | 12,344            | 12,344            | 12,344            | 12,344            |
| <b>EPS (\$)</b>                            |                  |                   |                   |                   |                   |                   |                   |                   |
| <b>Basic EPS</b>                           |                  |                   | -0.21             | 1.36              | 9.37              | 10.73             | 17.38             | 23.40             |
| <b>Diluted EPS</b>                         |                  |                   | -0.20             | 1.30              | 8.95              | 10.24             | 16.60             | 22.35             |
| <b>Adjusted EBITDA</b>                     | <b>\$ 5,350</b>  | <b>\$ 6,584</b>   | <b>\$ 21,206</b>  | <b>\$ 85,583</b>  | <b>\$ 180,205</b> | <b>\$ 233,754</b> | <b>\$ 339,093</b> | <b>\$ 486,986</b> |
| Space                                      | 1,154            | 653               | -561              | 2,817             | 3,122             | 2,939             | 3,768             | 4,556             |
| Connectivity                               | 3,849            | 7,168             | 10,520            | 17,075            | 30,947            | 53,498            | 78,205            | 114,737           |
| AI                                         | 347              | -1,237            | 11,247            | 65,690            | 146,136           | 177,318           | 257,120           | 367,694           |

Source: Company report, Bernstein analysis and estimates

EXHIBIT 18: **SpaceX Quarterly Income Statement**

| SpaceX Quarterly Income Statement (\$m)        |                  |                 |                   |                   |                   |                   |                   |                 |                  |                  |                   |
|------------------------------------------------|------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-----------------|------------------|------------------|-------------------|
| Fiscal Year Ending December 31                 |                  |                 |                   |                   |                   |                   |                   |                 |                  |                  |                   |
|                                                | 2024             |                 |                   |                   |                   | 2025              |                   |                 |                  |                  | 2026E             |
|                                                | 12 mo            | Q1              | Q2                | Q3                | Q4                | 12 mo             | Q1                | Q2              | Q3               | Q4               | 12 mo             |
| Space                                          | 3,796            | 865             | 746               | 1,666             | 809               | 4,086             | 619               | 962             | 1,538            | 1,675            | 4,794             |
| Connectivity                                   | 7,599            | 2,475           | 2,588             | 3,096             | 3,228             | 11,387            | 3,257             | 4,291           | 4,448            | 5,040            | 17,037            |
| AI                                             | 2,620            | 727             | 737               | 814               | 923               | 3,201             | 818               | 2,561           | 6,713            | 14,479           | 24,571            |
| <b>Total Revenues</b>                          | <b>\$ 14,015</b> | <b>\$ 4,067</b> | <b>\$ 4,071</b>   | <b>\$ 5,576</b>   | <b>\$ 4,960</b>   | <b>\$ 18,674</b>  | <b>\$ 4,694</b>   | <b>\$ 7,814</b> | <b>\$ 12,699</b> | <b>\$ 21,195</b> | <b>\$ 46,402</b>  |
| Cost of revenue                                | (7,996)          | (1,962)         | (2,282)           | (2,706)           | (2,501)           | (9,451)           | (2,388)           | (3,495)         | (4,591)          | (6,693)          | (17,167)          |
| Research & Development                         | (3,464)          | (1,557)         | (1,958)           | (2,291)           | (2,837)           | (8,643)           | (3,514)           | (3,548)         | (5,490)          | (10,649)         | (23,201)          |
| Selling, General, and administrative           | (1,813)          | (493)           | (606)             | (575)             | (970)             | (2,644)           | (746)             | (912)           | (1,091)          | (1,561)          | (4,309)           |
| Restructuring charges                          | (213)            | (4)             | (189)             | (278)             | (16)              | (487)             | 11                | (2)             | -                | -                | 9                 |
| Impairment                                     | (63)             | (24)            | (6)               | 1                 | (9)               | (38)              | -                 | -               | -                | -                | -                 |
| <b>Total cost and expenses</b>                 | <b>-13,549</b>   | <b>-4,040</b>   | <b>-5,041</b>     | <b>-5,849</b>     | <b>-6,333</b>     | <b>-21,263</b>    | <b>-6,637</b>     | <b>-7,957</b>   | <b>-11,172</b>   | <b>-18,903</b>   | <b>-44,669</b>    |
| Space                                          | 21               | -70             | -370              | 258               | -475              | -657              | -662              | -542            | -319.3394        | -523.6986        | -2047.038         |
| Connectivity                                   | 2006             | 1033            | 924               | 1189              | 1277              | 4423              | 1188              | 1656            | 1756.5306        | 1481.5922        | 6082.1228         |
| AI                                             | (1,561)          | (936)           | (1,524)           | (1,720)           | (2,175)           | (6,355)           | (2,469)           | (1,257)         | 90               | 1,334            | (2,302)           |
| <b>Operating income (loss) from operations</b> | <b>\$ 466</b>    | <b>\$ 27</b>    | <b>\$ (970)</b>   | <b>\$ (273)</b>   | <b>\$ (1,373)</b> | <b>\$ (2,589)</b> | <b>\$ (1,943)</b> | <b>\$ (143)</b> | <b>\$ 1,527</b>  | <b>\$ 2,292</b>  | <b>\$ 1,734</b>   |
| Interest income (expense)                      | (1,209)          | (330)           | (312)             | (359)             | (452)             | (1,453)           | (451)             | (289)           | (694)            | (694)            | (2,129)           |
| Other income (expense), net                    | 985              | (211)           | 413               | 140               | (519)             | (177)             | (1,876)           | (86)            | -                | -                | (1,962)           |
| <b>Income (loss) before income taxes</b>       | <b>\$ 242</b>    | <b>\$ (514)</b> | <b>\$ (869)</b>   | <b>\$ (492)</b>   | <b>\$ (2,344)</b> | <b>\$ (4,219)</b> | <b>\$ (4,270)</b> | <b>\$ (518)</b> | <b>\$ 833</b>    | <b>\$ 1,598</b>  | <b>\$ (2,357)</b> |
| Provisions for (benefit from) income taxes     | 549              | (14)            | (138)             | (562)             | (4)               | (718)             | (6)               | (23)            | -                | (53)             | (82)              |
| <b>Net income</b>                              | <b>\$ 791</b>    | <b>\$ (528)</b> | <b>\$ (1,007)</b> | <b>\$ (1,054)</b> | <b>\$ (2,348)</b> | <b>\$ (4,937)</b> | <b>\$ (4,276)</b> | <b>\$ (541)</b> | <b>\$ 833</b>    | <b>\$ 1,544</b>  | <b>\$ (2,439)</b> |
| <b>Shares outstanding (mn)</b>                 |                  |                 |                   |                   |                   |                   |                   |                 |                  |                  |                   |
| Basic                                          |                  |                 |                   |                   |                   |                   | 4,496             | 5,864           | 11,789           | 11,789           | 11,789            |
| Diluted                                        |                  |                 |                   |                   |                   |                   | 5,068             | 5,864           | 12,344           | 12,344           | 12,344            |
| <b>EPS (\$)</b>                                |                  |                 |                   |                   |                   |                   |                   |                 |                  |                  |                   |
| Basic EPS                                      |                  |                 |                   |                   |                   |                   | -\$0.95           | -\$0.09         | \$0.07           | \$0.13           | -\$0.21           |
| Diluted EPS                                    |                  |                 |                   |                   |                   |                   | -\$0.84           | -\$0.09         | \$0.07           | \$0.13           | -\$0.20           |
| <b>Adjusted EBITDA</b>                         | <b>5,350</b>     | <b>1,730</b>    | <b>1,212</b>      | <b>2,168</b>      | <b>1,474</b>      | <b>6,584</b>      | <b>1,127</b>      | <b>3,538</b>    | <b>6,800</b>     | <b>9,741</b>     | <b>21,206</b>     |
| Space                                          | 1,154            | 224             | -94               | 672               | -149              | 653               | -351              | -205            | 80               | -85              | -561              |
| Connectivity                                   | 3,849            | 1,618           | 1,583             | 1,921             | 2,046             | 7,168             | 2,087             | 2,597           | 2,799            | 3,037            | 10,520            |
| AI                                             | 347              | -112            | -277              | -425              | -423              | -1,237            | -609              | 1,146           | 3,921            | 6,789            | 11,247            |

Source: Company report, Bernstein analysis and estimates

EXHIBIT 19: **SpaceX Annual Balance Sheet**

| <b>SpaceX Annual Balance Sheet (\$m)</b>        |               |               |                |                |                |                |                  |                  |
|-------------------------------------------------|---------------|---------------|----------------|----------------|----------------|----------------|------------------|------------------|
| <b>Fiscal Year Ending December 31</b>           | <b>2024</b>   | <b>2025</b>   | <b>2026E</b>   | <b>2027E</b>   | <b>2028E</b>   | <b>2029E</b>   | <b>2030E</b>     | <b>2031E</b>     |
| Cash and cash equivalents                       | 11,385        | 24,747        | 63,003         | 59,295         | 128,410        | 145,412        | 236,854          | 225,752          |
| Marketable securities                           | 800           | 0             | 6,487          | 6,487          | 6,487          | 6,487          | 6,487            | 6,487            |
| Account receivables (net allowance credit loss) | 1,052         | 1,579         | 3,941          | 8,240          | 18,761         | 27,483         | 39,100           | 40,545           |
| Inventory                                       | 2,003         | 2,416         | 4,341          | 7,504          | 5,680          | 4,764          | 4,724            | 3,824            |
| Prepaid expenses and other current assets       | 868           | 2,210         | 2,550          | 4,035          | 3,234          | 3,040          | 3,265            | 2,777            |
| <b>Total current assets</b>                     | <b>16,108</b> | <b>30,952</b> | <b>80,322</b>  | <b>85,560</b>  | <b>162,572</b> | <b>187,186</b> | <b>290,431</b>   | <b>279,386</b>   |
| Property, plant, and equipment, net             | 21,147        | 42,602        | 116,519        | 328,948        | 550,165        | 801,617        | 1,074,408        | 1,345,960        |
| Finance lease right-of-use assets               | 1,686         | 1,260         | 892            | 533            | 0              | 0              | 0                | 0                |
| Intangible assets, net                          | 2,211         | 1,548         | 0              | 0              | 0              | 0              | 0                | 0                |
| Digital assets                                  | 1,749         | 1,637         | 0              | 0              | 0              | 0              | 0                | 0                |
| Goodwill                                        | 11,129        | 11,809        | 0              | 0              | 0              | 0              | 0                | 0                |
| Other assets                                    | 2,336         | 2,130         | 16,390         | 16,007         | 1,836          | 9,700          | 17,685           | 25,737           |
| <b>Total non current assets</b>                 | <b>40,258</b> | <b>60,986</b> | <b>133,801</b> | <b>345,488</b> | <b>552,001</b> | <b>811,317</b> | <b>1,092,093</b> | <b>1,371,697</b> |
| <b>Total assets</b>                             | <b>56,366</b> | <b>91,938</b> | <b>214,123</b> | <b>431,048</b> | <b>714,572</b> | <b>998,503</b> | <b>1,382,524</b> | <b>1,651,083</b> |
| Accounts payable                                | 4,413         | 11,792        | 8,217          | 18,606         | 21,653         | 28,836         | 37,691           | 41,355           |
| Deferred revenue, current                       | 10,179        | 12,116        | 27,099         | 43,007         | 32,469         | 31,782         | 42,724           | 30,449           |
| Accrued expenses and other liabilities          | 1,508         | 2,569         | 8,286          | 12,531         | 9,127          | 8,712          | 9,449            | 8,045            |
| <b>Total current liabilities</b>                | <b>16,100</b> | <b>26,477</b> | <b>43,602</b>  | <b>74,145</b>  | <b>63,249</b>  | <b>69,331</b>  | <b>89,864</b>    | <b>79,849</b>    |
| Debt                                            | 12,262        | 21,659        | 38,700         | 209,288        | 408,001        | 550,241        | 699,647          | 693,188          |
| Finance Lease Liabilities                       | 1,531         | 1,237         | 752            | 393            | 0              | 0              | 0                | 0                |
| Deferred Tax Liability / (Asset)                | -725          | -113          | 8              | 84             | -14,305        | -5,136         | 4,032            | 13,201           |
| Other Long-Term Liabilities                     | 1,394         | 1,353         | 1,293          | 1,293          | 1,293          | 1,293          | 1,293            | 1,293            |
| <b>Total non-current liabilities</b>            | <b>14,462</b> | <b>24,136</b> | <b>40,753</b>  | <b>211,058</b> | <b>394,989</b> | <b>546,398</b> | <b>704,973</b>   | <b>707,682</b>   |
| <b>Total liabilities</b>                        | <b>30,562</b> | <b>50,613</b> | <b>84,355</b>  | <b>285,203</b> | <b>458,238</b> | <b>615,729</b> | <b>794,837</b>   | <b>787,532</b>   |
| Convertible Preferred Stock                     | 20,941        | 38,752        | 0              | 0              | 0              | 0              | 0                | 0                |
| Additional Paid-in Capital                      | 35,868        | 37,710        | 167,344        | 167,344        | 167,344        | 167,344        | 167,344          | 167,344          |
| Retained Earnings                               | -31,005       | -35,137       | -37,576        | -21,498        | 88,991         | 215,430        | 420,343          | 696,207          |
| <b>Total shareholders' equity</b>               | <b>25,804</b> | <b>41,325</b> | <b>129,768</b> | <b>145,846</b> | <b>256,335</b> | <b>382,774</b> | <b>587,687</b>   | <b>863,551</b>   |
| <b>Total liabilities and equity</b>             | <b>56,366</b> | <b>91,938</b> | <b>214,122</b> | <b>431,048</b> | <b>714,573</b> | <b>998,503</b> | <b>1,382,524</b> | <b>1,651,082</b> |
| <b>Total Debt</b>                               | <b>13,793</b> | <b>22,896</b> | <b>39,452</b>  | <b>209,681</b> | <b>408,001</b> | <b>550,241</b> | <b>699,647</b>   | <b>693,188</b>   |
| <b>Net debt</b>                                 | <b>2,408</b>  | <b>-1,851</b> | <b>-23,552</b> | <b>150,386</b> | <b>279,590</b> | <b>404,829</b> | <b>462,793</b>   | <b>467,436</b>   |

Source: Company report, Bernstein analysis and estimates

EXHIBIT 20: **SpaceX Annual Cash Flow**

| <b>SpaceX Annual Cash Flow (\$m)</b>                                 |                    |                    |                    |                     |                     |                     |                     |                     |
|----------------------------------------------------------------------|--------------------|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| <b>Fiscal Year Ending December 31</b>                                | <b>2024</b>        | <b>2025</b>        | <b>2026E</b>       | <b>2027E</b>        | <b>2028E</b>        | <b>2029E</b>        | <b>2030E</b>        | <b>2031E</b>        |
| Net income (loss)                                                    | 791                | (4,937)            | (2,439)            | 16,078              | 110,489             | 126,439             | 204,913             | 275,864             |
| Depreciation & Amortization                                          | 3,824              | 6,701              | 16,586             | 54,611              | 63,819              | 78,703              | 103,716             | 145,687             |
| Stock Based Compensation                                             | 784                | 1,947              | 2,896              | 3,284               | 3,779               | 4,189               | 4,647               | 5,077               |
| Impairment & Other Charges                                           | 63                 | 38                 | -                  | -                   | -                   | -                   | -                   | -                   |
| Non-Cash Interest Income and Other                                   | (1,359)            | 947                | 1,433              | (703)               | (1,031)             | (330)               | (1,340)             | (1,179)             |
| Accounts Receivable                                                  | (347)              | (543)              | (2,362)            | (4,299)             | (10,521)            | (8,722)             | (11,617)            | (1,445)             |
| Inventory                                                            | (309)              | (413)              | (1,925)            | (3,163)             | 1,825               | 915                 | 40                  | 900                 |
| Prepaid Expenses & Other                                             | (328)              | (673)              | (340)              | (1,485)             | 800                 | 194                 | (225)               | 488                 |
| Accounts Payable                                                     | 472                | 709                | (3,575)            | 10,390              | 3,047               | 7,183               | 8,855               | 3,665               |
| Deferred Revenue                                                     | 1,876              | 1,929              | 14,983             | 15,908              | (10,538)            | (686)               | 10,941              | (12,274)            |
| Other Liabilities                                                    | 309                | 1,080              | 5,717              | 4,245               | (3,404)             | (415)               | 737                 | (1,405)             |
| <b>Net cash provided by operating activities</b>                     | <b>\$ 5,776</b>    | <b>\$ 6,785</b>    | <b>\$ 30,973</b>   | <b>\$ 94,866</b>    | <b>\$ 158,264</b>   | <b>\$ 207,469</b>   | <b>\$ 320,666</b>   | <b>\$ 415,377</b>   |
| Purchase of PPE                                                      | (11,163)           | (20,619)           | (90,502)           | (267,040)           | (285,036)           | (330,155)           | (376,507)           | (417,238)           |
| Other investing activities, net                                      | 367                | 1,044              | (7,206)            | -                   | -                   | -                   | -                   | -                   |
| <b>Net cash used in investing activities</b>                         | <b>\$ (10,796)</b> | <b>\$ (19,575)</b> | <b>\$ (97,708)</b> | <b>\$ (267,040)</b> | <b>\$ (285,036)</b> | <b>\$ (330,155)</b> | <b>\$ (376,507)</b> | <b>\$ (417,238)</b> |
| Debt Issuance / (Paydown)                                            | (77)               | 9,131              | 17,041             | 170,588             | 198,712             | 142,241             | 149,406             | (6,459)             |
| Finance Lease Payments                                               | (154)              | (295)              | (374)              | (470)               | (393)               | -                   | -                   | -                   |
| Equity Issuance / (Repurchases) **                                   | 12,061             | 17,514             | 88,746             | (1,653)             | (2,432)             | (2,553)             | (2,123)             | (2,781)             |
| <b>Net cash provided by financing activities</b>                     | <b>\$ 11,830</b>   | <b>\$ 26,350</b>   | <b>\$ 105,413</b>  | <b>\$ 168,465</b>   | <b>\$ 195,887</b>   | <b>\$ 139,687</b>   | <b>\$ 147,284</b>   | <b>\$ (9,241)</b>   |
| Effect of exchange rate changes on cash and cash equivalents         | 1                  | 63                 | (42)               | -                   | -                   | -                   | -                   | -                   |
| <b>Net change in cash and cash equivalents and restricted cash</b>   | <b>\$ 6,811</b>    | <b>\$ 13,623</b>   | <b>\$ 38,635</b>   | <b>\$ (3,708)</b>   | <b>\$ 69,115</b>    | <b>\$ 17,002</b>    | <b>\$ 91,443</b>    | <b>\$ (11,102)</b>  |
| <b>Cash and cash equivalents and restricted cash, beginning year</b> | <b>\$ 4,690</b>    | <b>\$ 11,501</b>   | <b>\$ 25,124</b>   | <b>\$ 63,759</b>    | <b>\$ 60,051</b>    | <b>\$ 129,166</b>   | <b>\$ 146,168</b>   | <b>\$ 237,610</b>   |
| <b>Cash and cash equivalents and restricted cash, end year</b>       | <b>\$ 11,501</b>   | <b>\$ 25,124</b>   | <b>\$ 63,759</b>   | <b>\$ 60,051</b>    | <b>\$ 129,166</b>   | <b>\$ 146,168</b>   | <b>\$ 237,610</b>   | <b>\$ 226,508</b>   |
| <b>FCF</b>                                                           | <b>(5,387)</b>     | <b>(13,834)</b>    | <b>(59,530)</b>    | <b>(172,174)</b>    | <b>(126,772)</b>    | <b>(122,685)</b>    | <b>(55,841)</b>     | <b>(1,861)</b>      |
| Conversion to net income %                                           | -681%              | 280%               | 2440%              | -1071%              | -115%               | -97%                | -27%                | -1%                 |
| <b>Working Capital</b>                                               | <b>(12,177)</b>    | <b>(20,272)</b>    | <b>(32,770)</b>    | <b>(54,366)</b>     | <b>(35,575)</b>     | <b>(34,044)</b>     | <b>(42,774)</b>     | <b>(32,703)</b>     |

Source: Company report, Bernstein analysis and estimates

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